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BATCH: T13 ROLL NO: 55

Explain software processing models in detail.

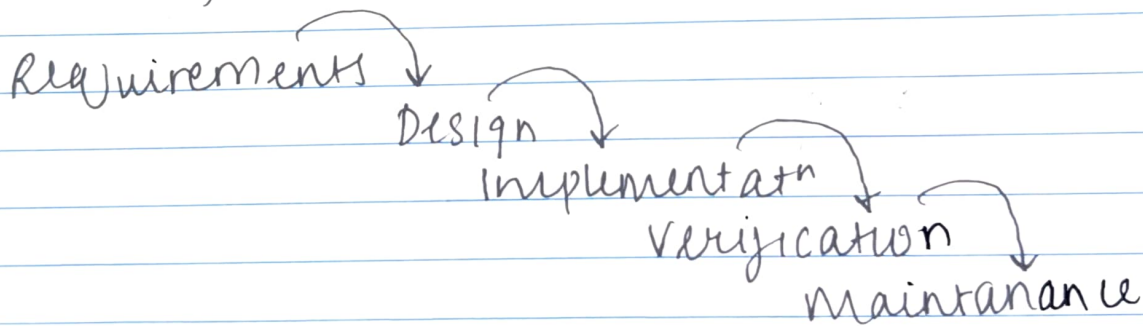
Software modeling:

definition:

Software modeling is the process of creating abstract representation of software system

These models serve as blueprint that guides developers, designers & stakeholders through system's structure, behaviour & functionality

A) Waterfall model:



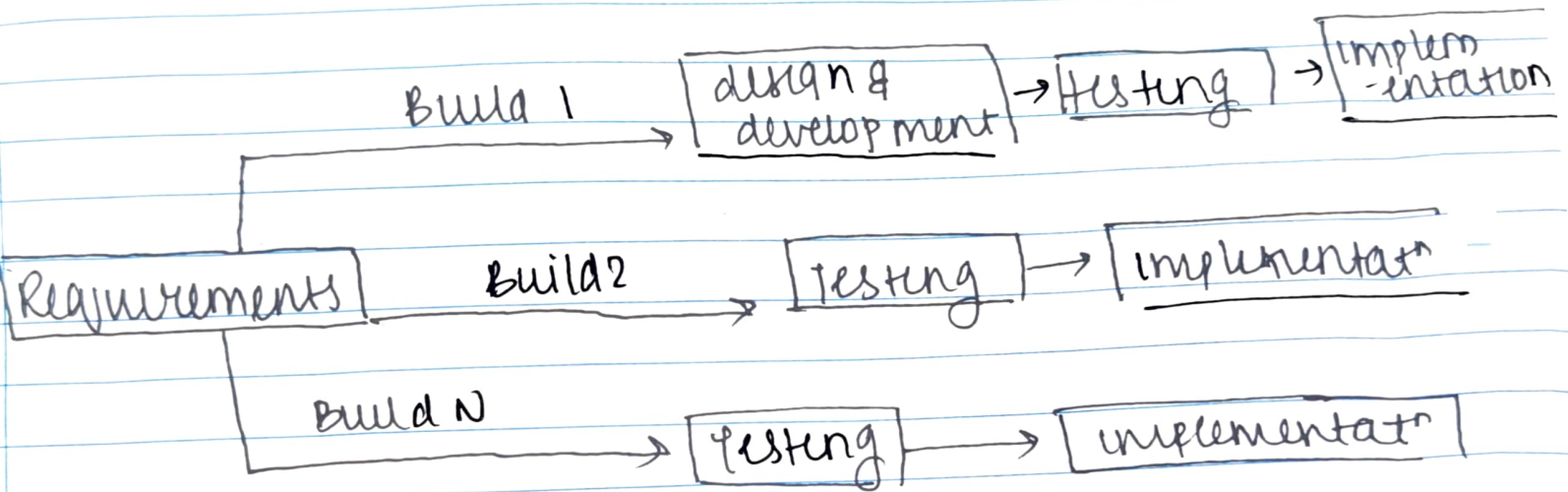
It is famous & good version of SDLC for SE. The waterfall model is linear & sequential model, which means that development phase cannot begin until previous phase is completed.

Adv: simple & easy to understand
well structured

Disadv: Not flexible to change
late testing phase increases risk

Best used when : Requirements are clear & stable
small & low risk projects.

(3) Incremental model



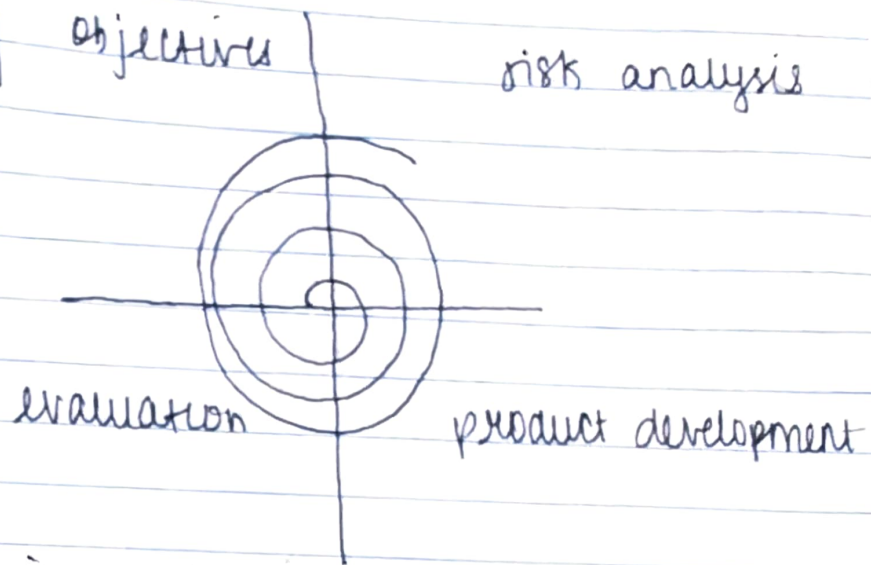
The incremental model develops the system in small parts (increments). Each increment adds new functionality until full system is completed.

Adv : Early delivery of partial system
easier testing & debugging

Disadv : Requires good planning & design
integration can be complex

Best used when system can be divided into

a) spiral model
identifying objectives



Spiral model is a risk-driven software development model that combines iteration development with systematic risk manage. It has 4 phases:

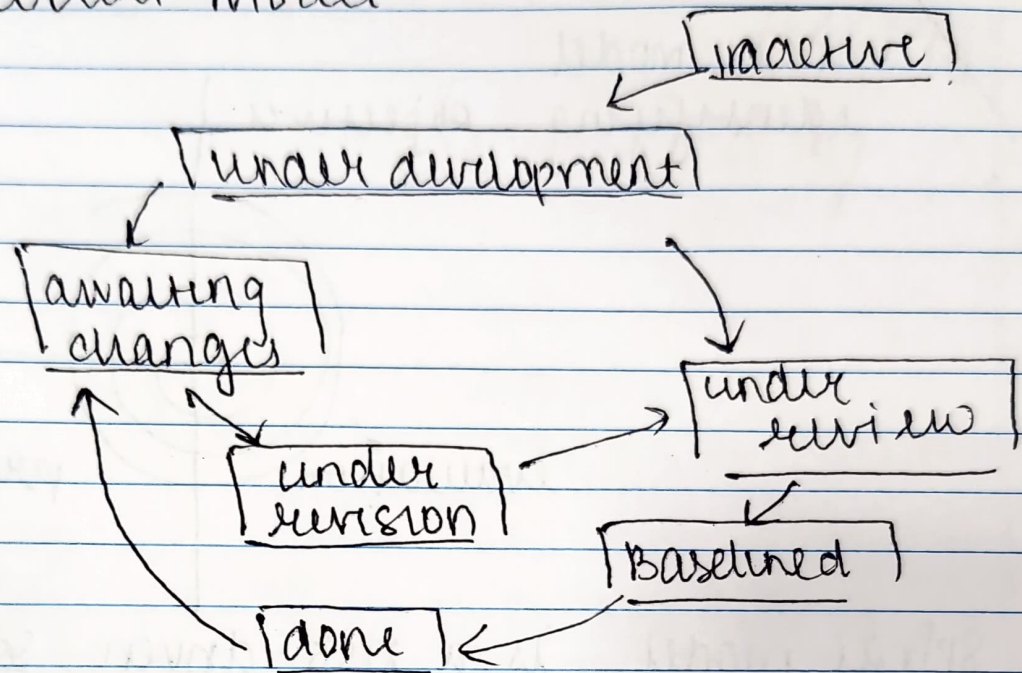
- 1] Planning : Define objects & alternatives
- 2] Risk analysis : Identify & migrate risks
- 3] Engineering : Design, code & test
- 4] Evaluation : Customer review & feedback

Adv : early identification of risks
flexible & suitable for complex to manage

Disadv : Expensive

Best used when : large, high-risk
new technology is involved

D) Concurrent model:



Concurrent development model allows multiple phases of development to occur simultaneously

Adv: faster development
Better handling of changes
Suitable for parallel work

Disadv: complex to manage
Requires effective communication

But used when:

large project with multiple teams
time to market is critical
system with frequent changes